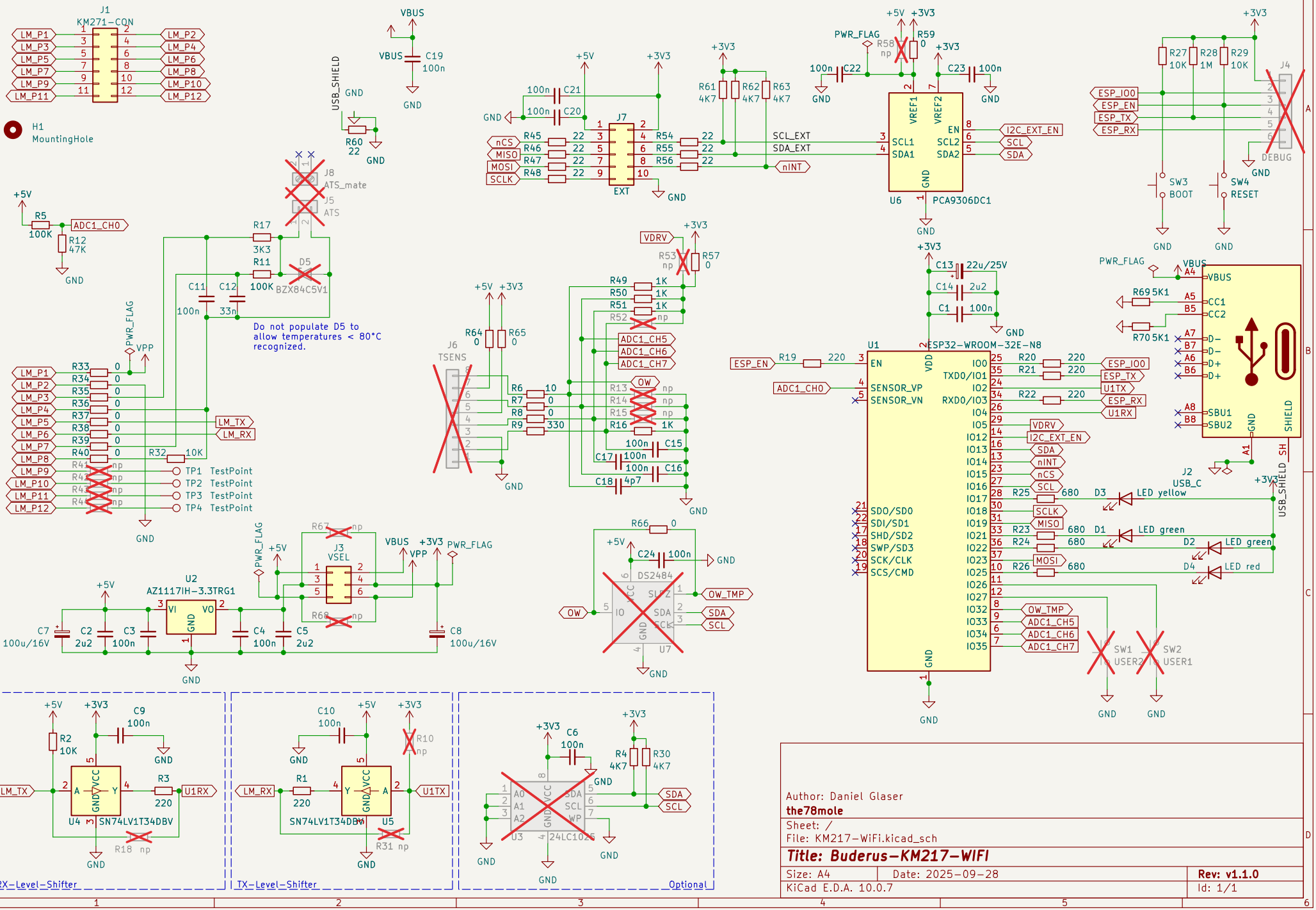

J1
KM271-CON
LM_P1
LM_P2
LM_P3
LM_P4
LM_P5
LM_P6
LM_P7
LM_P8
LM_P9
LM_P10
LM_P11
LM_P12
H1
MountingHole
VBUS
C19
100n
USB_SHIELD
R60
22
J7
EXT
nCS
MISO
MOSI
SCLK
R45
R46
R47
R48
R54
R55
R56
C21
C20
R61
R62
R63
4K7
SCL_EXT
SDA_EXT
nINT
PWR_FLAG
R58
R59
C22
C23
U6
PCA9306DC1
VREF1
VREF2
SCL1
SDA1
SCL2
SDA2
EN
I2C_EXT_EN
SCL
SDA
R27
R28
R29
10K
1M
J4
DEBUG
ESP_IO0
ESP_EN
ESP_TX
ESP_RX
SW3
BOOT
SW4
RESET
J8
ATS_mate
J5
ATS
R5
100K
R12
47K
ADC1_CH0
R17
3K3
R11
100K
D5
BZX84C5V1
C11
100n
C12
33n
Do not populate D5 to allow temperatures < 80°C recognized.
J6
TSENS
R64
R65
R49
R50
R51
R52
1K
ADC1_CH5
ADC1_CH6
ADC1_CH7
R6
R7
R8
R9
10
330
R13
R14
R15
R16
OW
C15
C16
C17
C18
4p7
VDRV
R53
R57
R33
R34
R35
R36
R37
R38
R39
R40
R41
R42
R43
R44
LM_TX
LM_RX
R32
TP1
TP2
TP3
TP4
TestPoint
U2
AZ1117IH-3.3TRG1
C7
100u/16V
C2
2u2
C3
C4
C5
C8
J3
VSEL
R67
R68
R66
DS2484
OW_TMP
SDA
SCL
C24
U7
U1
ESP32-WROOM-32E-N8
ESP_EN
R19
220
ADC1_CH0
SENSOR_VP
SENSOR_VN
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
IO0
TXD0/IO1
IO2
RXD0/IO3
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
R20
R21
R22
U1TX
U1RX
R25
680
R23
R24
R26
D3
LED yellow
D1
LED green
D2
LED green
D4
LED red
C13
22u/25V
C14
2u2
C1
100n
R69
R70
5K1
J2
USB_C
SBU1
SBU2
SHIELD
SW1
USER2
SW2
USER1
RX-Level-Shifter
R2
10K
R3
220
U4
SN74LV1T34DBV
R18
np
LM_TX
U1RX
TX-Level-Shifter
C10
R10
R1
U5
R31
LM_RX
U1TX
Optional
C6
R4
R30
4K7
U3
24LC1025
SDA
SCL
Author: Daniel Glaser
the78mole
Sheet: /
File: KM217-WiFi.kicad_sch
Title: Buderus-KM217-WIFI
Size: A4
Date: 2025-09-28
Rev: v1.1.0
KiCad E.D.A. 10.0.7
Id: 1/1